

370M people search Google Photos each month, and each one finds the photo only if four steps all work

1.5B+

people use Google Photos each month

9T+

photos and videos stored

370M+

people search their photos each month (1 in 4)

150M

Google One subscribers paying for storage

The business goal

"Increase the percentage of users who successfully retrieve a photo they remember but cannot precisely describe." A library people can't search is worth less to keep, and storage is what Google One sells.



It becomes one number: User Retrieval Rate (URR)

Of the people who search for a photo they can only vaguely describe, the share who reach it within 28 days. It counts people, not searches.

$$\text{URR} = E \times [1 - (1 - I \times S \times R)^n]$$

E Express it

Can they put the memory into words at all?

Today: a search box, or a question to Ask Photos

I Be understood

Does Photos read the clue the way they meant it?

Today: matches people, places and things

S See it surface

Is the photo in the results at all?

Today: a grid of matching photos

R Recognise it

Can they tell it is the right one?

Today: thumbnails with date and place

n Try again

How many tries before they give up?

Today: retype the search

How to read it: a person succeeds only if they can express the memory and one try passes I, S and R together. One weak step caps the whole number, so the next slides find which step breaks.

85,140 public posts became 144 comparable failed searches, showing what was kept, what was lost and where search broke

1 · COLLECT

85,140

Public posts
(Play Store,
App Store, YT,
Reddit)

2 · SCREEN

819

Relevant (Gate
A 5,305 → Gate
B 1,333
sample)

3 · EXTRACT

144

Specific
attempts (720
episodes; 62
scoreable)

4 · AUDIT

203

Pairs checked
blind by Qwen
27B model

5 · COMPARE

9 Areas

Ranked only on
agreed fields
(O1–O9)

Extracted Field in Schema	The Brief's Underlying Question Answered	Reliability Check
asset_type	What kinds of old photos do users struggle to retrieve?	Validated in Gate B
cues_retained (14 cue types)	What do people actually remember about the photo?	Jaccard 0.557
cues_lost (date, place, album, words)	What have they forgotten when search fails?	Jaccard 0.673
query_verbatim · search_mode	How do they search when memory is incomplete?	Quote verify 85.2%
failure_stage (5 stages)	Where does retrieval break? (Part 3 decomposition)	κ 0.509 (Moderate)

The audit, not the model, set the ranking

1. **Structure, not sentiment:** Every review was converted into a structured, typed record with fixed vocabularies, not qualitative impression tags.

2. **Pre-registered rules:** Hypotheses H1–H5 and decision thresholds were set before extraction, so the data could not be read to fit them.

3. **The audit overruled the ranking:** When the blind audit showed hypothesis agreement was low (Jaccard 0.347) while failure stage agreement was solid (κ 0.509), the engine demoted hypothesis ranking and pivoted to failure stages.

retrieval-discovery-engine.vercel.app

People keep "roughly when" and lose the exact date, the one thing search needs

Engine and survey agree: the time is kept, the date is lost

Question	Engine (144 Specific Attempts)	Survey (n=15 / 14 Dedup)
Photo Kinds	Personal photos 80 · multi-photo 26 · videos 16 · screenshots 7 · docs 3	All 3 real-trouble cases were document or medicine photos
Remembered	Roughly when 40 · object 17 · exact date 14 · text 12 · people 10	Roughly when 8 · object 7 · people with me 6
Forgotten	The date 37 · album 29 · exact words 10 · place 10	When it was taken 9 · words to search 9
Search Method	20 of 32 queries are 1 word ("dog"); 10 name a time ("Halloween 2024")	First move: scroll 9/15; own-app: 4 of 5 strip time to bare nouns

O1 is the largest group with a memory search ignores

O1 (Rough time or an event): 42 → 37 after exclusions (18 never surfaced, 12 misread).

O8 (No clue at all): 47 attempts, larger, but there is no memory for search to use. O1 is the largest group where memory exists and search ignores it.

Excluded: O7 exact date known (14) · O9 path moved by an update (12).

Episodic time wins; dead-end recovery loses

H1 Episodic time: SUPPORTED (root cause).

Roughly when is the #1 kept cue (40); the date is the #1 lost cue (37).

H2 Recognition: SUPPORTED (secondary). 41.7% never surfaced; 7 of 15 ended unsure.

H3 Dead-end recovery: REFINED, NOT THE LEAD. Overruled by the audit: recovery is 0.7% of failures (stage κ 0.509).

H4 Hinglish: NOT TESTED. Weak signal (2 of 15 survey, 1 of 6 tests); kept as a design choice.

H5 Text not indexed: WEAKLY SUPPORTED (3 docs; the index scores 0.000 on text in images).

H6 Path changed: PRESENT BUT MINOR (8.3%), added post hoc.

77% of the 144 failures break at I or S, before a retry

S Never surfaced	60 (41.7%)
I Clue misread	51 (35.4%)
E · R · retry	2 · 2 · 1

Most people scroll before they search, and nearly half end close but unsure

A structured survey gave 15 stories we could score

Structured self-serve questionnaire (n=15, 23–30 Sep): every answer maps onto the engine's vocabulary.

Why: 100% scoreable stories (vs 8.6% of public posts); reached 15 people vs 2 call contacts.

Costs: self-reported, convenience sample; 8 of 15 fit the segment.

[Survey questions ↗](#)

The survey repeats the engine's pattern

Metric	Engine (144)	Survey (15 / 14)
Most remembered	Roughly when (40)	Roughly when (8)
Ended unsure	Not measurable	7 of 15 (47%)
First move: scroll	15 workarounds	9 of 15 (60%)

It costs most people over five minutes

Trips and outings, a photo for LinkedIn, a medical report.

12 of 15 spent 5+ minutes, past the brief's five-minute bar; 3 had to ask someone or get the document again.

4 of 5 kept roughly when; only 1 put it in the search

ID	Target Photo	What They Remembered	What They Typed	Break	Outcome
S04	Wedding photo	Roughly when, event, people	"wedding, pichle saal diwali"	Interpretation	Unsure
S03	Personal photo	Roughly when, event, object	"gym"	Recognition	Unsure
S08	Personal photo	Roughly when, event	(Didn't know what to type; scrolled)	Expression	Found
S12	Personal photo	Roughly when, event	(Never typed; opened album)	Expression	Unsure
S06	Medicine bill	Object, text in bill	"medicine, bill"	Surfacing	Never found

"wedding, pichle saal diwali" — S04 (Hinglish: a festival and relative year, not a calendar date)

"medicine, bill" — S06 (Never found; had to get the document again)

Ask Photos: 7 of 15 had never heard of it; all 4 who used it got "related photos, but not the one".

Our segment can place the moment only roughly, and search can't turn "roughly" into dates

Defined by what they remember, not by age
Long-time Google Photos users searching for a personal moment they can place only roughly.

Defined by **cognitive memory state** (what they remember), not demographics: "around Diwali", "my sister's graduation", "last winter".

Segment Sizing Filter	Engine (144)	Survey (15)
Kept rough time or event (O1)	42 (29%)	8 (53%)
Target Segment (After Exclusions)	37 (26%)	8 (53%)
Personal photo or video	30 of 37	All 8
Large library (5,000+ items)	Not measured	5 of 8

42 → 37: minus 2 who also knew the exact date (plain search works) and 3 whose browse path moved in an update (H6). Sync and corruption issues are out of scope.

Memory stores episodes; Photos stores items and dates



"I'll type in something super simple, like 'Halloween 2024' and it seriously can't find anything?" — Play Store review

- Asking why, for the 37 segment attempts:
- 1. **Why isn't the photo found?** It never surfaces (18 of 37).
 - 2. **Why not?** Search misreads the clue (12): "last Diwali" is neither a keyword nor a date.
 - 3. **Why does that matter?** People keep the event and lose the date (the #1 lost cue, 37 of 144).
 - 4. **Root cause:** nothing turns "roughly when" into a date window.
Real app: 4 of 5 testers typed bare nouns ("vacation", "gym"); 0 of 5 found the photo.

People already scroll to "roughly when" by hand and still end unsure, so the product should take that step for them

What users can't do, and why

"When people want a photo they can place only roughly ("last Diwali"), they can describe the moment but can't make Photos show it: search needs a date or keyword they've forgotten. So they scroll by hand, and nearly half end unsure."

The Workaround is the Design Blueprint

Engine: 15 of 21 workarounds are manual scrolling (7 of 10 failed): *"Every time I scrolled to the right time frame, it kept loading more pictures..."*

Survey: Scrolling was the first move for 9 of 15. 5 of those 9 still ended unsure.

Insight: people already retrieve by moment, by hand.
The product should find the time window and group the moments.

Google already holds the data this needs

Option	How you search
Apple Photos (iOS 18)	Describe what's in the photo
Ask Photos	Ask Gemini; all 7 who tried got "related, not the one"
Memory Trails	Roughly when → date window → moments, with why

Why Google can win: it already holds every user's dates, places and faces, and could read events from Calendar. No standalone app has that data.

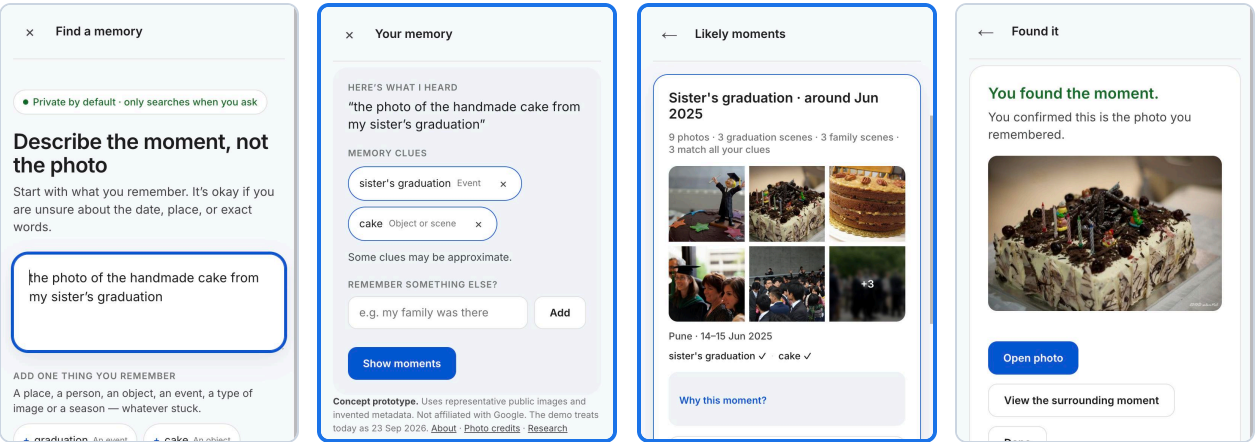
Five times the evidence changed the plan

1. Metric: per-search success → per-user URR	2. Outcomes: recovery agent → 5 stages + browse	3. Discovery: H3 ranked top → overturned by audit	4. Reported behaviour: retry search → scroll by hand	5. Problem: "search is bad" → items vs. moments
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Memory Trails turns "last Diwali" into a date window and shows moments, not a grid

It sits inside search, where the failure happens

Where it lives: inside Photos search. When a search fails, "Can't describe it?" opens it with the words carried over.



1 · Describe 2 · Clues 3 · Moments 4 · Confirm

memory-trails-v2.vercel.app Blue frame = AI step

- Four mechanisms behind the gain
- 1. Clues → date window: "pichle saal Diwali"
 - 2. Moments, not a grid: "3 match all your clues"
 - 3. "Why this moment?": matched vs approximate
 - 4. Soft scoring: a wrong date lowers rank, never hides

Big gain with rich memory, modest with one vague cue

Cue Dropout Level	What the Query Keeps	Plain CLIP	Memory Trails	Moment@5
L3 (Fully described)	Vague time + exact place + library word	0.172	0.962	0.933
L2 (Two vague cues)	Paraphrased content + 2 vague cues	0.209	0.276	0.333
L1 (One vague cue)	Paraphrased content + 1 vague time cue	0.242	0.309	0.333
L0 (Content only)	Pure scene description, no metadata	0.312	0.312	0.267
Weighted by real memory	Each level weighted by how often the 144 real attempts keep it	0.259	0.334 (+0.075)	0.336 (+0.111)

Read this first: L3's 0.962 needs three cues; only 4% (6 of 144) of real attempts kept that many. Weighted by real memory, recall gains +0.075 and moment@5 +0.111: grouping into moments helps more than flat ranking.

Scope: tests the re-entry interaction on a 1,282-photo Creative Commons library, not production-scale accuracy.

5 of 6 testers were sure they found the photo, but "year before last" still breaks it

ID	Own Photos search → result	Task 1 (cat)	Task 2 (dog)	Sure	vs GP	Would use
R01	"Wedding before COVID" → wrong years	Found, sure	Not found	5/5	5/5	Every time
R02	"gym" → can't tell which	Found, sure	Found (1-tap alt)	4/5	4/5	When fails
R03	(Apple Photos user)	Found, sure	Found (nearby)	5/5	—	Every time
R04	"vacation" → wrong years	Found, sure	Found (1-tap alt)	4/5	4/5	When fails
R05	"Wedding event" → can't tell which	Found, sure	Found (clue edit)	4/5	4/5	Every time
R06	"Kerala photos" → can't tell which	Right day, unsure	Found (1-tap alt)	4/5	3/5	Every time

6 of 6

Fit the segment

4.33 / 5

Task 1 sureness

4.0 / 5

Ease vs Photos

0 of 5

Found it in own Photos

MVP test questions ↗

memory-trails-v2.vercel.app ↗

Testers valued the moment, not a grid

"Shows the moment, not a grid" — R02

"Moving to nearby months, like scrolling but faster" — R03

"Works without knowing what to type" — R04

"Finds exact month & year... understands mix of Hindi and English" — R06

Three fixes for the next version

1. Compound time: R01 stuck on "pichle ke pichle saal" (year before last). Fix: parse compound offsets; editable year.

2. Clues missed: R04 didn't notice them. Fix: high-contrast "Showing Diwali 2025 · Tap to change".

3. Speed: R05, R06 wanted faster loading. Fix: pre-cache moments.

Success is URR rising from 40% to 45% in a user-level A/B test, with an early signal per step

One number: users who reach the photo in 28 days
Definition: share of users with a vague-memory search in 28 days who reach the photo. Counts users, not searches.
Baseline 40% → target 45%. The baseline is a self-reported proxy (6 of 15 sure they found it). +5 pts is the smallest lift worth shipping.
Test: 8 weeks, randomised by user; ~1,600 users per arm for 80% power at α 0.05.

Recall and confidence already moved on the MVP
Recall@20 weighted by real memory **0.259 → 0.334** · moment@5 **+0.111** · tester sureness **4.33 / 5** · ease vs Photos **4.0 / 5**

Ship, extend or stop is decided in advance
Ship if URR rises ≥ 5 pts and no guardrail trips. **Extend** if the lift is positive but under 5. **Stop** if false confirmation passes 3%.

Each URR term gets an early signal

Metric	Term	Target
Entry rate after a failed search	Expression	$\geq 12\%$
Clue correction (chip edited)	Interpretation	15–25%
"Not in any of these?" rate	Surfacing	Falling
Confirm after opening a moment	Recognition	$\geq 55\%$

A wrong 'found it' stops the launch

False confirmation (confirmed, then kept searching)	$< 3\%$
P95 time to first moment	< 1.5 s
Abandonment before the first moment	No rise vs control

Diagnostics explain any movement
Recall@20 by cue type · near-miss recovery rate · "Why this moment?" views before confirming

Biggest risk: the gain is largest where memory is richest, and the evidence is still small

Each risk has a mitigation built or planned

Risk	Mitigation & evidence
R1. The win sits where memory is richest (+0.79 at L3, +0.07 at L1)	Lean on moment recognition (moment@5 0.333 at L1–L2); report L1 separately in the A/B.
R2. Date filters hide photos	Soft scoring built: L3 recall 0.851 → 0.962.
R3. False confirmation (worst failure)	Never auto-confirm; show the surrounding photos; stop launch above 3%.
R4. Can't read text in photos (0.000 on receipts)	Add OCR text to the index.
R5. Ask Photos may already suffice	4 of 4 survey and 3 of 3 test users got "related photos, not the one".
R6. Sensitive memories (medical, relationships)	User-initiated only; the session persists nothing.

The evidence is small and mostly self-reported

- Play Store is 86.2% of engine posts
- Survey n = 15 vs a target of 30, self-reported
- MVP tests n = 6, unmoderated, no session logs
- URR baselines modelled; no telemetry
- Eval library: 1,282 CC photos, synthetic episodes
- Ranking weights are judgement
- Hinglish (H4) is a weak signal only

Next: compound dates, clearer clues, a one-market test

1. Parse compound dates ("pichle ke pichle saal")
2. Make the active date clue unmissable
3. Run the A/B in one market